Carlos Ojeda-de-Silva

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Education

Harvey Mudd College August 2023 – May 2027
Bachelor of Science in Engineering (GPA: 3.5 / 4.0) *Claremont, CA*

Experience

Lead, Electrical and Computer Engineer August 2023 – Present
HMC Liquid Rocket Engine *Claremont, CA*

- Manufactured a liquid rocket engine designed to produce 1,000+ lbf of thrust.
- Lead the electrical and software team for the Liquid Rocket Engine.
- Architected electrical systems for telemetry and propellant feed control on a liquid rocket test stand.

Biomedical Engineer January 2025 – Present
CuTE Lab at Harvey Mudd College *Claremont, CA*

- Created a 4-channel pressure pump for precise microfluidic control.
- Retrofitted a 3D printer for pressure extrusion of polyethylene glycol based bioinks.
- Developed a custom toolpath slicer to print arbitrary geometries while optimizing extrusion paths to minimize extruded material stiffness gradient.
- Engineered a PDMS microfluidic device with specialized geometric features to control individual channel resistance and optimize fluid mixing.

Electrical and Computer Engineer August 2026 – Present
Blackrock Neurotech *Claremont, CA*

- Engineer a custom ESP32 board to compress and relay BCI data over Wi-Fi with minimal power draw.
- Evaluate power consumption metrics to ensure the system meets continuous multi-hour runtime requirements for neuro-telemetry.

Mechanical Engineer August 2025 – December 2025
Northstar Radioisotopes *Claremont, CA*

- Engineered specialized maintenance tools for robotic systems to minimize worker radiation exposure.
- Designed and tested a dual-function tool to disconnect pipe connectors and cut steel pipes with minimal burrs.

Mechanical Engineer November 2023 – December 2024
Sokil Imaging *Claremont, CA*

- Engineered and validated lightweight mounts for magnetometers reducing mechanical oscillations and signal noise.
- Patented drone sensor mounting currently deployed in Ukraine for de-mining through magnetometry.

Recovery Engineer August 2021 – June 2023
Alamo Heights High School Rocketry Program *San Antonio, TX*

- Manufactured custom rocket components and conducted detailed trajectory analysis to safely recover a 140-lb rocket from 50,000 ft.
- Presented flight simulation data to NASA engineers and launched a supersonic vehicle at White Sands Missile Range, achieving an altitude of 18,994 ft.

Projects

Compressed Video Streaming | *Jupyter Notebook, ESP32, LoRa* May 2024 – Present

- Reduce LoRa bandwidth requirements by implementing neural network autoencoder compression
- Evaluate other data compression algorithms (PCA, Frequency Domain Compression) to maximize bandwidth.
- Architect a pipeline to send Accelerometer, Barometer, and GPS data through the LoRa channel.

Portable DJ Set | *Questa, Lattice Radiant, SEGGER* October 2025 – December 2025

- Implemented real-time audio processing at dynamic sampling by interfacing an MCU with an FPGA for DSP.
- Designed a joystick controller to change playback speed and applied FIR filtering.

Technical Skills

Languages: Java, Python, C/C++, System Verilog, MATLAB
Developer Tools: SolidWorks, Altium Designer, VS Code, Git, Lattice Radiant, Questa, Google Colab, Jupyter Notebook, Arduino IDE, SEGGER
Additional Skills: CAD, PCB Design, Machining, Single Board Computer, Microcontroller, FPGA, 3D printing, Laser Cutting, Finite Element Analysis